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Defect Detection in an Open-Set Setting via Multi-Bin Test-Time Adaptation

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Introduction

- Actual wafer manufacturing easily breaks down existing models due to covariate shifts and semantic shifts[2,3].
- Conventional Test-Time Adaptation (TTA) methods assume only in-distribution (ID) data, leading to severe performance degradation when Out-Of-Distribution(OOD) data is mixed[4,5,6,7].
- Wafermap data is sparse, highly structured, and multimodal, making conventional approaches unsuitable[2,3].
- Proposal: 1) Mahalanobis distance-based scoring for covariate-shifted In-Distribution (csID)/OOD separation, 2) Multi-bin histogram segmentation to model complex wafer distributions, 3) Selective weight updates to suppress degradation caused by OOD
- Objective: Improve OOD separation while maintaining csID accuracy, enabling stable open-set defect detection in industrial environments.

Related Works

- Open-Set Recognition (OSR) is the problem of distinguishing new classes not present in training, with active research on distance-based methods and energy/entropy-based scoring.
- Existing TTA approaches exhibit an open-set limitation where models are easily contaminated when OOD data is mixed during updates, failing to properly account for the impact of OOD data's mere presence.
- Recent studies propose methods that first distinguish between csID and OOD data before selectively updating, but they suffer from the limitation of insufficiently capturing subtle distribution differences.
- This research builds upon the ideas of these prior techniques but proposes an improved approach better suited to the characteristics of industrial wafer data.

Methods

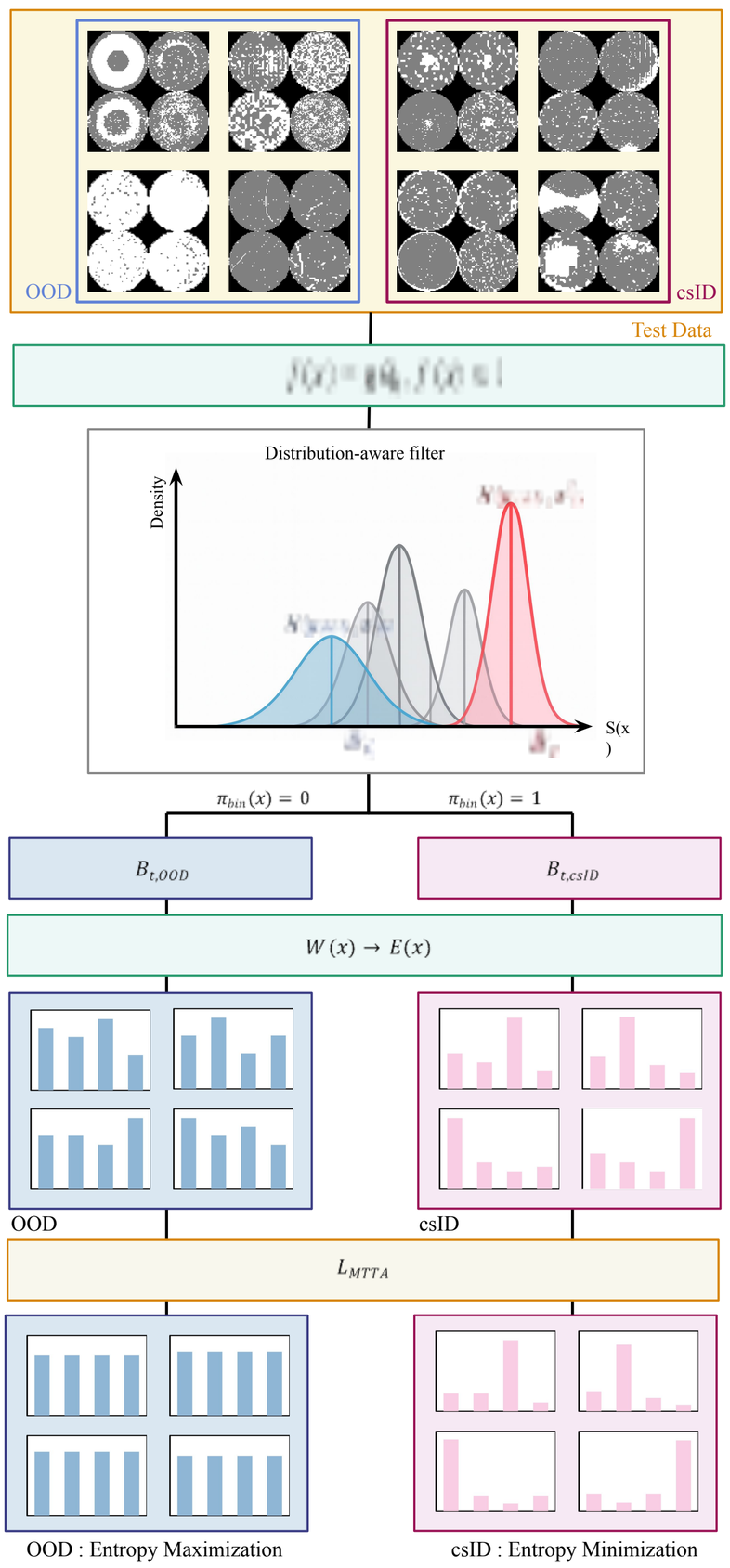

Figure 1. Overall Structure of Multi-bin Test-Time Adaptation (MTTA)

- The model operates based on a pre-trained feature extractor using a ResNet-18 backbone.
- Test data is input with a batch size of 256, and each sample is transformed into a high-dimensional embedding vector through the backbone network.
- The Mahalanobis distance S(x) is calculated based on statistical information such as class centers and covariance to measure each sample's deviation from its class center.
- Samples close to csID yield small negative values, while those close to OOD yield large negative values.
- Scores are normalized via min-max normalization, then divided into intervals using a multi-bin structure to determine the bin each sample belongs to.
- The leftmost bin is selected as the OOD candidate, and the rightmost bin as the csID candidate.
- A trimming strategy is applied within each bin, selecting only the top and bottom 30% of samples to ensure only low-noise samples are used for updates.
- Positive weights are assigned to csID samples and negative weights to OOD samples, structuring the loss to minimize and maximize entropy, respectively.
- The final TTA loss is defined as the sum of these weighted entropy losses.
- Only the affine parameters (γ, β) of BatchNorm are updated via gradient descent, ensuring the model adapts stably during the testing phase.

Results

- Measurement Environment: Open-set environment using WM-811K data.
- Source Model: Refers to the baseline model without TTA (ResNet-18).
- Existing TTA Methods: Mistakenly identified OOD samples as ID, leading to incorrect updates and degraded performance[6,10].
- MTTA(Multi-Bin Test-Time Adaptation) : Achieved higher open-set performance than the source model thanks to Mahalanobis score-based separation and multi-interval selective updates.
- MTTA significantly improved OOD discrimination while nearly maintaining ID performance.
- Distance distributions and t-SNE visualizations also demonstrated that MTTA formed clearer boundaries between csID and OOD, proving stable adaptation effects.
- Overall, MTTA's multi-interval distribution separation and weighted entropy strategy were confirmed to suppress over-adaptation—a core issue in open-set TTA—and improve csID/OOD separation.

Method	Total Acc	AUROC	FPR@TPR95	OSCR
source	95.84%	70.29%	78.68%	68.86%
Tent [6]	92.07%	30.98%	97.07%	27.63%
Tent + unient [10]	87.08%	28.87%	97.94%	23.81%
Tent + unient+ [10]	90.49%	35.55%	95.66%	31.12%
MTTA	**96.09%**	**75.91%**	**77.34%**	**74.51%**

Table 1. This table evaluates the open-set adaptation performance of the source, existing TTA (Tent/UniEnt), and the proposed MTTA in the WM-811K open-set environment.

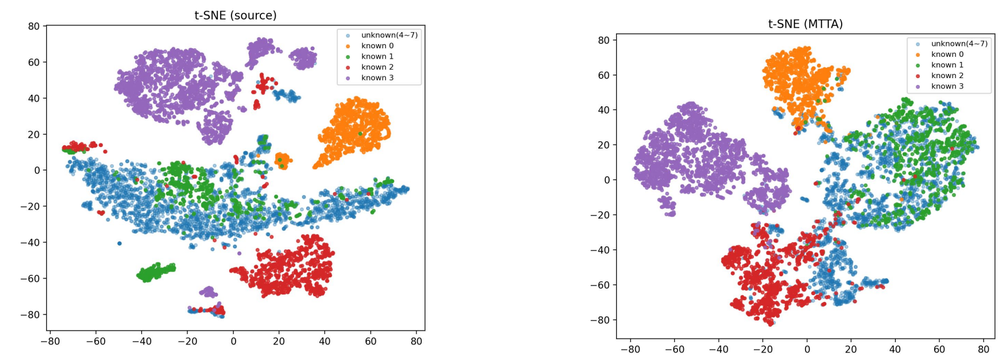

Figure 2. Clear separation of csID and OOD clusters in t-SNE after MTTA application..

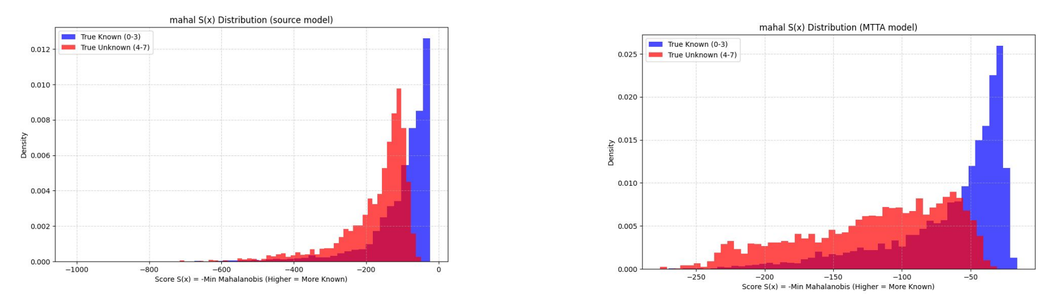

Figure 3. Improved separation of Mahalanobis-based S(x) score distribution after MTTA application.

Conclusion

- This study proposes an MTTA technique that combines Mahalanobis distance-based scores reflecting wafer map characteristics, multi-bin distribution separation, and selective weight updates.
- This significantly enhances OOD separation capability while maintaining csID performance, addressing the issue where existing TTA incorrectly updates OOD samples in open-set environments, leading to performance degradation.
- However, this study is specialized for the WM-811K dataset, limiting its generalizability to other domains. Due to MTTA's fully unsupervised learning nature, performance limitations exist compared to partially labeled data environments [4,5,6].
- The multi-bin separation approach may be sensitive to batch composition and bin count selection. Therefore, additional strategies for optimizing the number of bins and automated bin determination methods are required based on the characteristics of the target data, the distribution structure between classes, and the nature of covariate/semantic shifts.
- Future research should focus on: verifying domain scalability across diverse manufacturing processes and general vision datasets; introducing semi-supervised TTA techniques utilizing partial label information; applying alternative distance-based metrics (e.g., energy score, prototype distance) beyond Mahalanobis distance; stability evaluation on Transformer-based backbone models beyond ResNet, and developing multi-bin dynamic update or adaptive weight learning strategies. This would enable the framework to evolve into a more universal open-set TTA solution.

Reference

- [2] M.-J. Wu, J.-S. R. Jang, and J.-L. Chen, “Wafer Map Failure Pattern Recognition and Similarity Ranking for Large-Scale Data Sets,” IEEE Transactions on Semiconductor Manufacturing, 28(1), pp. 1–12, 2015.
- [3] L. Frittoli, D. Carrera, B. Rossi, P. Fragneto, and G. Boracchi, “Deep Open-Set Recognition for Silicon Wafer Production Monitoring,” Pattern Recognition, 2022.
- [4] Z. Nado, S. Padhy, D. Sculley, A. D’Amour, B. Lakshminarayanan, and J. Snoek, “Evaluating Prediction-Time Batch Normalization for Robustness under Covariate Shift,” arXiv:2006.10963, 2020.
- [5] S. Niu et al., “Efficient Test-Time Model Adaptation Without Forgetting,” in ICML, 2022.
- [6] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, and T. Darrell, “TENT: Fully Test-Time Adaptation by Entropy Minimization,” in ICLR, 2021.
- [10] Z. Gao, X.-Y. Zhang, and C.-L. Liu, “Unified Entropy Optimization for Open-Set Test-Time Adaptation,” in CVPR, 2024.